LIM-1.A.1 Given a function f, the limit of f(x) as x approaches c is a real number R if f(x) can be made arbitrarily close to R by taking x sufficiently close to c (but not equal to c). If the limit exists and is a real number, then the common notation is $\lim_{x \to c} f(x) = R$.
LIM-1.B.1 A limit can be expressed in multiple ways, including graphically, numerically, and analytically.
LIM-1.C.1 The concept of a limit includes one sided limits.
LIM-1.C.2 Graphical information about a function can be used to estimate limits.
LIM-1.C.3 Because of issues of scale, graphical representations of functions may miss important function behavior.
LIM-1.C.4 A limit might not exist for some functions at particular values of x. Some ways that the limit might not exist are if the function is unbounded, if the function is oscillating near this value, or if the limit from the left does not equal the limit from the right.

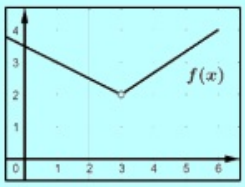
1-2 & 1-3 Limit Notation and Graphical Limits

The foundation of all calculus is the concept of a **LIMIT**.

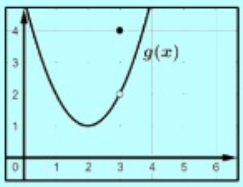
We need to understand several ideas about limits:

- What is a limit?
- How is a limit similar/different than a point on the graph?
- How do we write limit statements?
- How do we read limit statements?

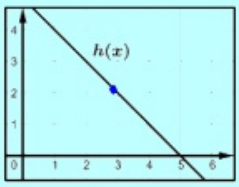
A limit is a y-value that a graph/function gets closer and closer to (but may not actually get to) as the x-values get closer and closer to a specific value from both sides.



$\lim_{x \to 3} f(x) = 2$



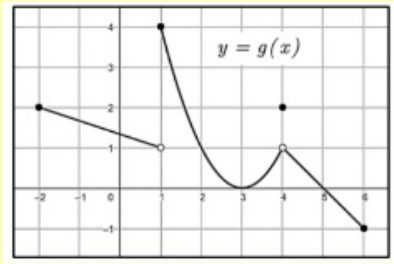
$\lim_{x \to 3} g(x) = 2$



$\lim_{x \to 3} h(x) = 2$

Note: All three graphs above have the same limit value as x approaches 3.

Recall from Precalculus: $\lim_{x \to 3} f(x) = 2$ is read as "The limit of $f(x)$ as x approaches 3 is 2"



<u>Example 1:</u> Use the graph of $g(x)$ above to answer the following:

a) $\lim_{x \to 4} g(x) =$ 1

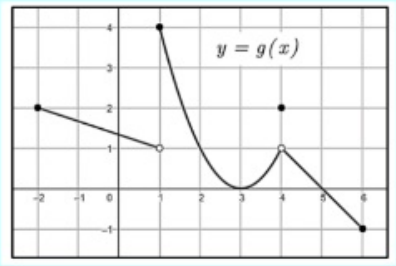
b) $\lim_{x \to 1} g(x) =$ DNE

c) $g(4) =$ 2

d) $\lim_{x \to 2} g(x) =$ 1

One Sided Limits

In Example 1, $\lim_{x \to 1} g(x)$ does not exist because we are approaching two different y values from each side of the graph. This brings up the idea of one-sided limits

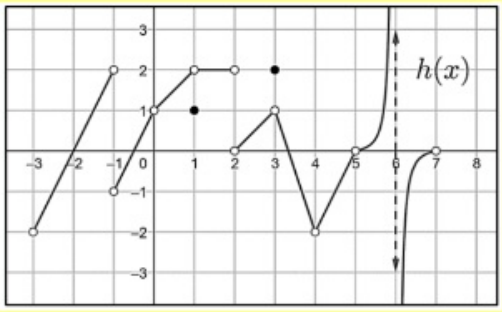


From the left: $\lim_{x \to 1^-} g(x) =$ 1 left-hand limit

From the right: $\lim_{x \to 1^+} g(x) =$ 4 right-hand limit

Be careful! From the left does NOT mean that you are moving left. It means START to the left of the x value but you will actually be moving towards the right.

Example 2: Use the graph of $h(x)$ above to answer the following:

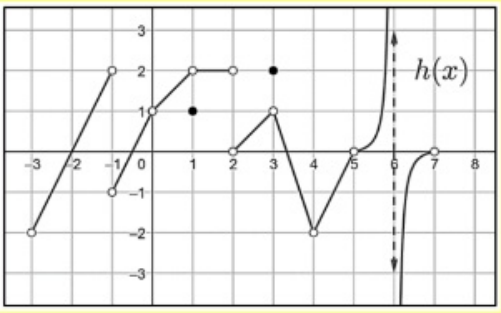


a) $\lim_{x \to 3} h(x) =$ 1

b) $\lim_{x \to 2} h(x) =$ DNE

$\lim_{x \to 2^-} h(x) \neq \lim_{x \to 2^+} h(x)$

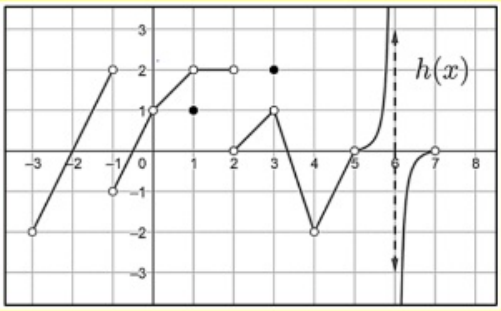
c) $\lim_{x \to 2^+} h(x) =$ 0



d) $h(1) =$ 1

e) $\lim_{x \to -1^-} h(x) =$ 2

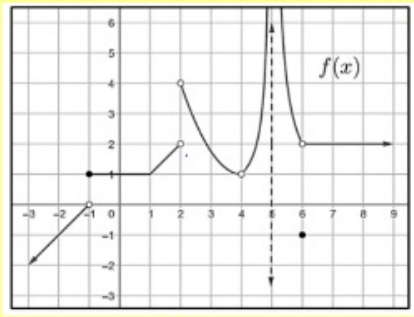
f) $\lim_{x \to -1^+} h(x) =$ −1



g) $\lim_{x \to -1} h(x) =$ DNE

h) $h(-1) =$ undefined

Example 3: Write a limit statement for each of the criteria below.



$\lim_{x \to 5^-} f(x) = +\infty$

a) A limit that exists: $\lim_{x \to 4} f(x) = 1$

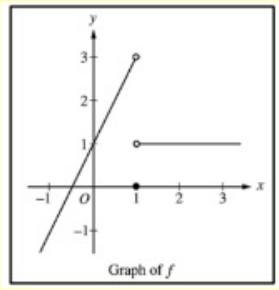
b) A left sided limit: $\lim_{x \to 2^-} f(x) = 2$

c) A right sided limit: $\lim_{x \to 2^+} f(x) = 4$

d) A limit that does not exist: $\lim_{x \to 2} f(x) = \text{DNE}$

Example 4: The graph of $y = f(x)$ is shown above. What is $\lim_{x \to 1} f(x)$?

(A) 0
(B) 1
(C) 3
(D) The limit does not exist



Congratulations! You have correctly answered your first AP Exam problem! The problem above was on the 2018 AP Exam (AB3)